TFY4235/FY8904: Computational Physics (semester YYYY)

Assignment 2: Signature Forge Detection (SFD)

Authors: Per Ivar Friborg, Raffaela Cabriolu

Department of Physics, NTNU

Abstract

Even though most documents today are digital, handwritten signatures are still commonly used to confirm identity, such as in contracts, banking, and official paperwork. At the same time, high-quality scanners and image editing tools make it easy to copy and manipulate signatures. This makes signature forgery detection (SFD) a practical and relevant problem, where computational methods must work with noisy data, limited samples, and subtle visual differences.

Starting from a mathematical description of grayscale images as discrete scalar fields, the assignment guides the student through preprocessing, feature extraction, optimization based alignment, and learning an embedding space in which genuine and forged signatures can be distinguished. The work culminates in training and evaluating a Siamese neural network for SFD, with particular emphasis on generalization, overfitting, and the geometric structure of learned embedding spaces.

Relevant fields: Signature Forge Detection (SFD), Machine Learning, Computer Vision.
Mathematical and numerical methods: Linear algebra, discretized scalar fields, convolution, Fourier transforms, numerical optimization, stochastic gradient descent, loss functions, embedding spaces, confusion matrices, dimensionality reduction (t-SNE).
Contact: Per Ivar Friborg (perivarfriborg@gmail.com), Raffaela Cabriolu (raffaela.cabriolu@ntnu.no).

1 Introduction

Signature forge detection (SFD) is a persistent challenge in banking, legal, and administrative institutions. In many realistic scenarios only static images of handwritten signatures are available, and no dynamic information such as writing speed or pen pressure can be accessed. The relevance of SFD has increased in recent years as digital copies of documents have become more common, making high quality reproductions and forgeries easier to distribute and harder to detect by visual inspection alone.

From a computational physics perspective, an image may be viewed as a discrete scalar field defined on a rectangular grid. The task of SFD can then be formulated as a numerical similarity problem, where two scalar fields representing grayscale images are mapped into a feature space in which distances reflect whether a pair of signatures is genuine or forged. Classical approaches relied on handcrafted geometric or statistical descriptors, but such methods are sensitive to intra class variability, meaning the natural variation between genuine signatures from the same individual. In this assignment we instead use machine learning models to learn these variations directly from data.

The well known CEDAR Signature Dataset is introduced here as background context. In its raw form it contains 2 640 grayscale signature images scanned at high resolution. To support efficient experimentation and reproducibility, this assignment provides a curated subset rather than the full corpus. The purpose of this curation is twofold, to substantially reduce memory and storage requirements, and to ensure that all students have a dataset that is immediately usable for building dataloaders and training models without additional preprocessing.

The signatures are provided in two parallel subsets. The first subset contains grayscale signature images, while the second contains binarized versions of the same signatures. In both cases, all images share a common resolution obtained through padding, which allows them to be loaded directly into neural network models without further resizing or shape handling.

In addition to the image files, a CSV file is provided to define the dataset structure. This file specifies the file path to each signature image, the corresponding writer identity, and a boolean label indicating whether the signature is genuine or a forgery. Students are expected to use this CSV file to construct their own datasets and dataloaders, and to control how signatures are grouped, paired, and split for training, validation, and testing in the context of signature forgery detection (SFD).

2 Images as Scalar Fields and Vectors

2.1 Grayscale Images as Discrete Fields

A grayscale image can be represented mathematically as a discrete scalar field

$$I(x, y), \qquad (x, y) \in \{0, 1, \ldots, H-1\} \times \{0, 1, \ldots, W-1\},$$

where $I(x, y)$ denotes the intensity at pixel (x, y).

a) Describe the same image equivalently as a matrix in $\mathbb{R}^{H \times W}$ and as a vector in $\mathbb{R}^{HW}$. Explain explicitly how you map between the scalar field, matrix, and vector representations.

b) Given two images I_1 and I_2 of the same size viewed as vectors in $\mathbb{R}^{HW}$, write down the Euclidean distance $\|I_1 - I_2\|_2$. Interpret this quantity in terms of pixel wise differences and discuss why it is a natural but limited similarity measure for signatures.

c) Pixel intensities are commonly stored either as 8 bit unsigned integers in $\{0, \ldots, 255\}$ or as floating point values rescaled to $[0, 1]$. Discuss how this choice affects numerical stability in operations such as averaging and normalization, and how it affects the interpretation of distances and inner products between images.

d) Read the provided grayscale image `exampleSignature.png` using `cv2.imread`. What is the resulting data type and what range of values do the pixel intensities take.

3 Pixel Intensity Distributions

Let $I(x, y)$ be a grayscale image with pixel intensities in $\{0, 1, \ldots, 255\}$.

a) Define the empirical probability mass function

$$p(k) = \mathbb{P}(\text{pixel intensity} = k), \qquad k = 0, \ldots, 255,$$

in terms of a sum over all pixel locations using indicator functions or Kronecker deltas.

b) Explain how a histogram with B bins approximates this distribution. Discuss the effect of choosing a small versus a large number of bins, and relate this to separating background from ink in a signature image. Visualize the histogram for two different values of B, using `exampleSignature.png`.

4 Image Inversion

Continue using `exampleSignature.png` for this task. Consider a normalized grayscale image $I(x, y) \in [0, 1]$. We define the inverted image by the explicit transformation

$$\mathcal{T}[I](x, y) = 1 - I(x, y).$$

a) View I as a vector in $\mathbb{R}^{HW}$. Is the transformation $\mathcal{T}$ linear. Justify your answer.

b) Describe how the histogram of pixel intensities changes under $\mathcal{T}$. What happens to the relative positions of background and ink intensities.

c) Suppose each image is subsequently normalized to zero mean and unit variance. Argue whether applying $\mathcal{T}$ before this normalization is likely to change the information available to a learning algorithm.

d) At a conceptual level, discuss whether image inversion is expected to make SFD easier or harder for a sufficiently expressive model.

5 Kernels and Convolution

A kernel is a small matrix used in convolution to combine local neighborhoods of pixels. Given an image I and a kernel K, the convolution output is

$$I'(x, y) = \sum_{i,j} K(i, j)\, I(x + i, y + j).$$

Consider the following kernels:

$$\text{A } \frac{1}{9}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}, \quad \text{B } \begin{bmatrix} 0 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 0 \end{bmatrix}, \quad \text{C } \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}.$$

a) Apply kernel A to a signature image and describe its effect.

b) Apply kernel B and describe how it modifies edges and fine details.

c) Apply kernel C and interpret the resulting image.

d) Design a kernel that emphasizes a specific geometric feature of your choice and explain its effect.

6 Thresholding

Convert a grayscale signature image into a binary image by thresholding. Choose a threshold that separates foreground ink from background effectively. Visualize the binary image and its histogram, and explain why thresholding is useful for segmentation and feature extraction.

7 Optimization Based Alignment

The vertical height of a signature image is defined here as the vertical pixel extent of the foreground after cropping away background regions. Consider rotating an image by an angle θ and then cropping background. In this assignment you must use `exampleSignature_diag.png`.

a) Define an objective function that measures the cropped image height as a function of θ.

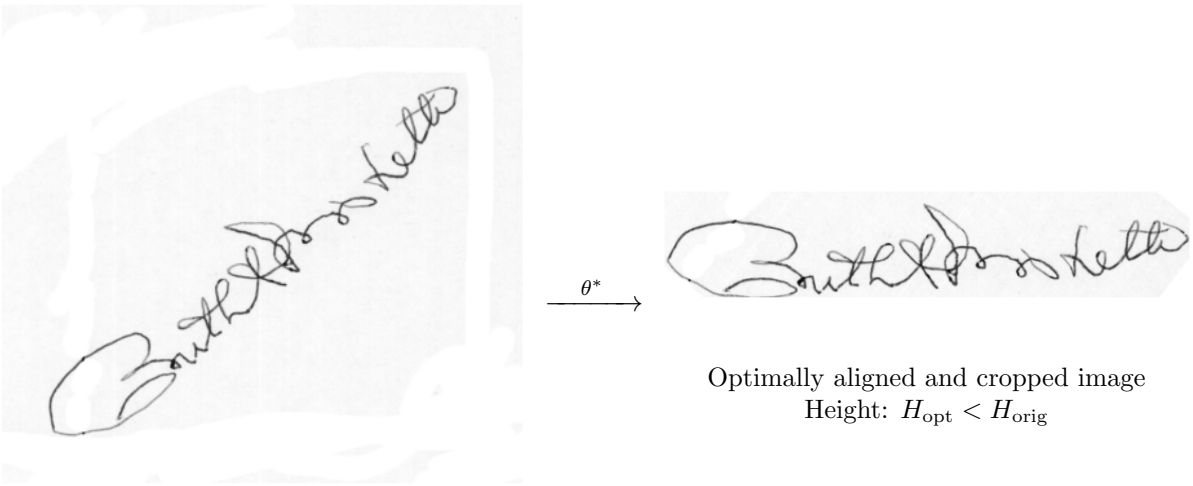
b) Use `cv2.warpAffine` to rotate the image.

c) Use a finite difference method to find the angle θ that minimizes the height.

d) Discuss the numerical stability of the finite difference method in this setting and how the choice of step size affects the result.

e) Visualize the original and optimally aligned images.

f) Discuss one advantage and one disadvantage of applying this preprocessing step for SFD.



Optimally aligned and cropped image
Height: $H_{\text{opt}} < H_{\text{orig}}$

Original signature image
Height: H_{orig}

Figure 1: Illustration of optimization-based alignment. The signature image is rotated by an angle θ and cropped to its foreground. The optimal angle θ^* minimizes the vertical height of the cropped image, which reduces the required padded image resolution and memory usage in downstream SFD models.

8 Fourier Based Denoising

a) Apply the two dimensional Fourier transform to a grayscale image.

b) Choose a frequency domain threshold to suppress high frequency noise and justify your choice graphically.

c) Apply the inverse transform and visualize the denoised image.

d) Compare the result to the original image and comment on any loss of detail.

9 Dataset Size, Filtering, and Generalization

The provided dataset differs from the original CEDAR dataset primarily through filtering rather than downscaling, since all images must be padded to a common resolution determined by the largest retained image.

a) Discuss how filtering out very large signature images, reducing the number of writers, and limiting the number of signatures per writer may affect model performance. In your discussion address generalization, overfitting, and the ability of the trained model to handle signatures that differ from those seen in the curated dataset.

10 From Images to Pairs and Batches

In SFD we work with pairs of signatures (S_1, S_2, y), where $y = 0$ denotes a genuine genuine pair and $y = 1$ denotes a genuine forged pair.

a) For a single grayscale image of shape (H, W), write down the tensor shape used in TensorFlow or Keras and in PyTorch. Explain what each dimension represents.

b) For a mini batch of size B, write down the shapes of the tensors representing the first image batch, the second image batch, and the labels.

c) Implement a small data loading pipeline that produces such batches and verify the shapes and data types.

11 Convolutional Neural Networks and Siamese Architecture

A convolutional neural network maps a signature image S to an embedding $e = f_\theta(S) \in \mathbb{R}^d$. To compare two signatures, we use a Siamese architecture consisting of two identical copies of this network with shared parameters.

Given a pair (S_1, S_2), the network produces embeddings e_1 and e_2, which are compared using a distance measure. A common choice is the Euclidean distance

$$D(e_1, e_2) = \|e_1 - e_2\|_2.$$

One way to train such a network is through the contrastive loss

$$L(y, D) = yD^2 + (1 - y)\max(0, m - D)^2,$$

where $y \in \{0, 1\}$ is the pair label and $m > 0$ is a margin parameter.

a) Propose a CNN architecture suitable for extracting signature embeddings.

b) Starting from the input image, track and report the spatial resolution and tensor dimensions after each convolution and pooling layer in your network.

c) Explain the role of the embedding layer and why embeddings are preferable to raw convolutional feature maps for SFD.

d) Interpret the contrastive loss and explain what it encourages for genuine genuine pairs and for genuine forged pairs.

e) Explain why the two branches of a Siamese network must share parameters.

12 Training, Evaluation, and Visualization

a) Choose an optimizer and training hyperparameters and justify your choices.

b) Train the network and plot training and validation curves.

c) Compute a confusion matrix on the test set and interpret the results in the context of SFD.

d) Visualize embeddings using t SNE before and after training and discuss what you observe.